



ELMALG: Polynomials and Factoring

Introduction to Statistics · C-ID ELMALG · numberbender.com

Name: _____

Date: _____

Score: / 30

Learning Objectives

- Apply the laws of exponents (product, quotient, power)
- Multiply polynomials using FOIL and the distributive property
- Factor polynomials using GCF and trinomial patterns
- Simplify polynomial expressions

Simplify each expression completely. Show all steps and circle your final answer.

Exponent rules

1. Simplify: $x^7 \cdot x^2$. Express as a single power of x .

$$x^7 \cdot x^2$$

Answer: _____

2. Simplify using the quotient rule: x^5 / x^2 (assume x is not 0)

$$x^5 \div x^2$$

Answer: _____

3. A scientist writes x^{11} / x^5 in her formula. Simplify to a single power.

$$x^{11} \div x^5$$

Answer: _____

4. Simplify: $x^4 \cdot x^2$. Express as a single power of x .

$$x^4 \cdot x^2$$

Answer: _____

5. Simplify using the quotient rule: x^8 / x^6 (assume x is not 0)

$$x^8 \div x^6$$

Answer: _____

6. A scientist writes x^8 / x^3 in her formula. Simplify to a single power.

$$x^8 \div x^3$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Polynomials & Factoring Worksheet — Elementary Algebra

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 1 of 11 · Video: youtu.be/EJFvomRXPkM



Watch the lesson



More free sheets

7. Simplify: $x^6 \cdot x^3$. Express as a single power of x .

$$x^6 \cdot x^3$$

Answer: _____

8. Simplify using the quotient rule: x^9 / x^5 (assume x is not 0)

$$x^9 \div x^5$$

Answer: _____

9. A scientist writes x^7 / x^5 in her formula. Simplify to a single power.

$$x^7 \div x^5$$

Answer: _____

Factoring out the GCF

10. Factor completely by pulling out the greatest common factor: $4x^2 + 32x$

$$4x^2 + 32x$$

Answer: _____

11. The area of a rectangle is $2x^2 + 18x$ square units. Factor the expression to find possible dimensions.

$$2x^2 + 18x$$

Answer: _____

12. Factor completely by pulling out the greatest common factor: $2x^2 + 10x$

$$2x^2 + 10x$$

Answer: _____

13. The area of a rectangle is $6x^2 + 18x$ square units. Factor the expression to find possible dimensions.

$$6x^2 + 18x$$

Answer: _____

14. Factor completely by pulling out the greatest common factor: $5x^2 + 30x$

$$5x^2 + 30x$$

Answer: _____



15. The area of a rectangle is $2x^2 + 14x$ square units. Factor the expression to find possible dimensions.

$$2x^2 + 14x$$

Answer: _____

Adding and subtracting polynomials

16. Add: $(3x^2 + 2x + 4) + (5x^2 + 8x + 5)$

$$(3x^2 + 2x + 4) + (5x^2 + 8x + 5)$$

Answer: _____

17. Subtract: $(7x^2 + 8x + 4) - (3x^2 + 5x + 2)$

$$(7x^2 + 8x + 4) - (3x^2 + 5x + 2)$$

Answer: _____

18. Add: $(1x^2 + 6x + 5) + (5x^2 + 1x + 2)$

$$(1x^2 + 6x + 5) + (5x^2 + 1x + 2)$$

Answer: _____

19. Subtract: $(7x^2 + 6x + 4) - (2x^2 + 1x + 2)$

$$(7x^2 + 6x + 4) - (2x^2 + 1x + 2)$$

Answer: _____

20. Add: $(1x^2 + 2x + 4) + (4x^2 + 7x + 3)$

$$(1x^2 + 2x + 4) + (4x^2 + 7x + 3)$$

Answer: _____

21. Subtract: $(8x^2 + 10x + 8) - (1x^2 + 5x + 2)$

$$(8x^2 + 10x + 8) - (1x^2 + 5x + 2)$$

Answer: _____

Multiplying polynomials (FOIL)

22. Multiply using FOIL: $(1x + 3)(2x + 1)$.

$$(1x + 3)(2x + 1)$$

Answer: _____



23. Multiply using FOIL: $(4x + 5)(3x + 4)$

$$(4x + 5)(3x + 4)$$

Answer: _____

24. Expand: $(x + 2)(x - 7)$.

$$(x + 2)(x - 7)$$

Answer: _____

25. Multiply using FOIL: $(1x + 1)(3x + 4)$.

$$(1x + 1)(3x + 4)$$

Answer: _____

26. Multiply using FOIL: $(3x + 5)(2x + 5)$

$$(3x + 5)(2x + 5)$$

Answer: _____

27. Expand: $(x + 4)(x - 2)$.

$$(x + 4)(x - 2)$$

Answer: _____

28. Multiply using FOIL: $(3x + 1)(1x + 2)$.

$$(3x + 1)(1x + 2)$$

Answer: _____

29. Multiply using FOIL: $(1x + 5)(4x + 8)$

$$(1x + 5)(4x + 8)$$

Answer: _____

30. Expand: $(x + 3)(x - 5)$.

$$(x + 3)(x - 5)$$

Answer: _____





ELMALG: Polynomials and Factoring

Introduction to Statistics · C-ID ELMALG · numberbender.com

ANSWER KEY & SOLUTIONS

Topics: Multiplying polynomials (FOIL), Factoring out the GCF, Adding and subtracting polynomials, Exponent rules. All answers verified by independent computation.

Solutions



Exponent rules

1. Simplify: $x^7 \cdot x^2$. Express as a single power of x .

$$x^7 \cdot x^2$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)} = x^9$.

Answer: x^9

2. Simplify using the quotient rule: x^5 / x^2 (assume x is not 0)

$$x^5 \div x^2$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

→ $m = 5$, $n = 2$. So $x^5 / x^2 = x^3$.

Answer: x^3

3. A scientist writes x^{11} / x^5 in her formula. Simplify to a single power.

$$x^{11} \div x^5$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$. Here $m = 11$, $n = 5$.

→ $x^{11} / x^5 = x^6$.

Answer: x^6

4. Simplify: $x^4 \cdot x^2$. Express as a single power of x .

$$x^4 \cdot x^2$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)} = x^6$.

Answer: x^6

5. Simplify using the quotient rule: x^8 / x^6 (assume x is not 0)

$$x^8 \div x^6$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

→ $m = 8$, $n = 6$. So $x^8 / x^6 = x^2$.

Answer: x^2

6. A scientist writes x^8 / x^3 in her formula. Simplify to a single power.

$$x^8 \div x^3$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$. Here $m = 8$, $n = 3$.

→ $x^8 / x^3 = x^5$.

Answer: x^5

7. Simplify: $x^6 \cdot x^3$. Express as a single power of x .

$$x^6 \cdot x^3$$

→ Product rule: $x^m \cdot x^n = x^{(m+n)} = x^9$.

Answer: x^9



8. Simplify using the quotient rule: x^9 / x^5 (assume x is not 0)

$$x^9 \div x^5$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$.

→ $m = 9$, $n = 5$. So $x^9 / x^5 = x^4$.

Answer: x^4

9. A scientist writes x^7 / x^5 in her formula. Simplify to a single power.

$$x^7 \div x^5$$

→ Quotient rule: $x^m / x^n = x^{(m-n)}$. Here $m = 7$, $n = 5$.

→ $x^7 / x^5 = x^2$.

Answer: x^2



Factoring out the GCF

10. Factor completely by pulling out the greatest common factor: $4x^2 + 32x$

$$4x^2 + 32x$$

→ Find the GCF of $4x^2$ and $32x$: $GCF = 4x$.

→ Factor it out: $4x^2 + 32x = 4x(x + 8)$.

Answer: $4x(x + 8)$

11. The area of a rectangle is $2x^2 + 18x$ square units. Factor the expression to find possible dimensions.

$$2x^2 + 18x$$

→ Factor out the GCF of $2x^2$ and $18x$: $GCF = 2x$.

→ Dimensions: $2x(x + 9)$.

Answer: $2x(x + 9)$

12. Factor completely by pulling out the greatest common factor: $2x^2 + 10x$

$$2x^2 + 10x$$

→ Find the GCF of $2x^2$ and $10x$: $GCF = 2x$.

→ Factor it out: $2x^2 + 10x = 2x(x + 5)$.

Answer: $2x(x + 5)$

13. The area of a rectangle is $6x^2 + 18x$ square units. Factor the expression to find possible dimensions.

$$6x^2 + 18x$$

→ Factor out the GCF of $6x^2$ and $18x$: $GCF = 6x$.

→ Dimensions: $6x(x + 3)$.

Answer: $6x(x + 3)$

14. Factor completely by pulling out the greatest common factor: $5x^2 + 30x$

$$5x^2 + 30x$$

→ Find the GCF of $5x^2$ and $30x$: $GCF = 5x$.

→ Factor it out: $5x^2 + 30x = 5x(x + 6)$.

Answer: $5x(x + 6)$

15. The area of a rectangle is $2x^2 + 14x$ square units. Factor the expression to find possible dimensions.

$$2x^2 + 14x$$

→ Factor out the GCF of $2x^2$ and $14x$: $GCF = 2x$.

→ Dimensions: $2x(x + 7)$.

Answer: $2x(x + 7)$



Adding and subtracting polynomials

16. Add: $(3x^2 + 2x + 4) + (5x^2 + 8x + 5)$

$$(3x^2 + 2x + 4) + (5x^2 + 8x + 5)$$

→ Combine like terms: $(3+5)x^2 = 8x^2$, $(2+8)x = 10x$, $(4+5) = 9$.

→ Result: $8x^2 + 10x + 9$.

Answer: $8x^2 + 10x + 9$

17. Subtract: $(7x^2 + 8x + 4) - (3x^2 + 5x + 2)$

$$(7x^2 + 8x + 4) - (3x^2 + 5x + 2)$$

→ Distribute the minus sign to the second polynomial.

→ Combine like terms: $(7-3)x^2$, $(8-5)x$, $(4-2)$.

→ Result: $4x^2 + 3x + 2$.

Answer: $4x^2 + 3x + 2$

18. Add: $(1x^2 + 6x + 5) + (5x^2 + 1x + 2)$

$$(1x^2 + 6x + 5) + (5x^2 + 1x + 2)$$

→ Combine like terms: $(1+5)x^2 = 6x^2$, $(6+1)x = 7x$, $(5+2) = 7$.

→ Result: $6x^2 + 7x + 7$.

Answer: $6x^2 + 7x + 7$

19. Subtract: $(7x^2 + 6x + 4) - (2x^2 + 1x + 2)$

$$(7x^2 + 6x + 4) - (2x^2 + 1x + 2)$$

→ Distribute the minus sign to the second polynomial.

→ Combine like terms: $(7-2)x^2$, $(6-1)x$, $(4-2)$.

→ Result: $5x^2 + 5x + 2$.

Answer: $5x^2 + 5x + 2$

20. Add: $(1x^2 + 2x + 4) + (4x^2 + 7x + 3)$

$$(1x^2 + 2x + 4) + (4x^2 + 7x + 3)$$

→ Combine like terms: $(1+4)x^2 = 5x^2$, $(2+7)x = 9x$, $(4+3) = 7$.

→ Result: $5x^2 + 9x + 7$.

Answer: $5x^2 + 9x + 7$

21. Subtract: $(8x^2 + 10x + 8) - (1x^2 + 5x + 2)$

$$(8x^2 + 10x + 8) - (1x^2 + 5x + 2)$$

→ Distribute the minus sign to the second polynomial.

→ Combine like terms: $(8-1)x^2$, $(10-5)x$, $(8-2)$.

→ Result: $7x^2 + 5x + 6$.

Answer: $7x^2 + 5x + 6$



Multiplying polynomials (FOIL)

22. Multiply using FOIL: $(1x + 3)(2x + 1)$.

$$(1x + 3)(2x + 1)$$

→ F: $2x^2$. O: $1x$. I: $6x$. L: 3 .

→ Combine: $2x^2 + 7x + 3$.

Answer: $2x^2 + 7x + 3$

23. Multiply using FOIL: $(4x + 5)(3x + 4)$

$$(4x + 5)(3x + 4)$$

→ First: $4x * 3x = 12x^2$.

→ Outer: $4x * 4 = 16x$. Inner: $5 * 3x = 15x$.

→ Last: $5 * 4 = 20$. Sum the middle terms: $31x$.

→ Result: $12x^2 + 31x + 20$.

Answer: $12x^2 + 31x + 20$

24. Expand: $(x + 2)(x - 7)$.

$$(x + 2)(x - 7)$$

→ FOIL: $x^2 - 7x + 2x - 2*7 = x^2 + -5x + -14$.

Answer: $x^2 - 5x - 14$

25. Multiply using FOIL: $(1x + 1)(3x + 4)$.

$$(1x + 1)(3x + 4)$$

→ F: $3x^2$. O: $4x$. I: $3x$. L: 4 .

→ Combine: $3x^2 + 7x + 4$.

Answer: $3x^2 + 7x + 4$

26. Multiply using FOIL: $(3x + 5)(2x + 5)$

$$(3x + 5)(2x + 5)$$

→ First: $3x * 2x = 6x^2$.

→ Outer: $3x * 5 = 15x$. Inner: $5 * 2x = 10x$.

→ Last: $5 * 5 = 25$. Sum the middle terms: $25x$.

→ Result: $6x^2 + 25x + 25$.

Answer: $6x^2 + 25x + 25$

27. Expand: $(x + 4)(x - 2)$.

$$(x + 4)(x - 2)$$

→ FOIL: $x^2 - 2x + 4x - 4*2 = x^2 + 2x - 8$.

Answer: $x^2 + 2x - 8$



28. Multiply using FOIL: $(3x + 1)(1x + 2)$.

$$(3x + 1)(1x + 2)$$

→ F: $3x^2$. O: $6x$. I: $1x$. L: 2 .

→ Combine: $3x^2 + 7x + 2$.

Answer: $3x^2 + 7x + 2$

29. Multiply using FOIL: $(1x + 5)(4x + 8)$

$$(1x + 5)(4x + 8)$$

→ First: $1x * 4x = 4x^2$.

→ Outer: $1x * 8 = 8x$. Inner: $5 * 4x = 20x$.

→ Last: $5 * 8 = 40$. Sum the middle terms: $28x$.

→ Result: $4x^2 + 28x + 40$.

Answer: $4x^2 + 28x + 40$

30. Expand: $(x + 3)(x - 5)$.

$$(x + 3)(x - 5)$$

→ FOIL: $x^2 - 5x + 3x - 3*5 = x^2 + -2x + -15$.

Answer: $x^2 - 2x - 15$

